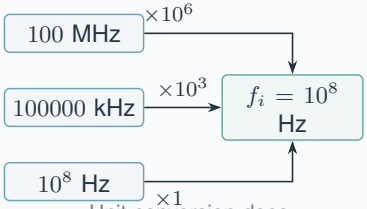
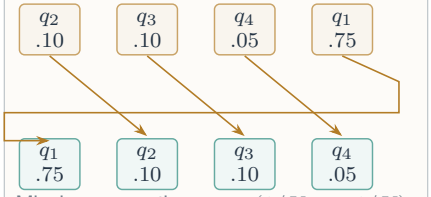
Observation contract and representation invariance

(a) Units



(b) Class order

Native order of receiver R_1



(c) Observable record

Identity: receiver s_i, position $\mathbf{r}_{s_i}$

Measurements: $t_i,\ f_i,\ B_i,\ D_i$

Uncertainty: measurement s.d.s; receiver residual bias declarations

Semantics: canonical vector π_i

Observed fields only; no event/source label



Shared biases make pairs statistically dependent.

Evaluation-only boundary

True event IDs, clutter flags and scene seeds never enter feature construction.